Quantum Field Theory – Homework 1

Name: ______________________ **Due Date:** 2025.10.27

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Classical Field Theories and the Principle of Locality (20 points)

Summarize the key concepts presented in **Lecture 1** on *The Principle of Locality and Fields* and **Lecture 2** on *Classical Field Theories*. Your summary should demonstrate a clear understanding of how these ideas motivate the formulation of modern field theories.

- Discuss the motivation for introducing field theory starting from classical mechanics.
- Explain the **principle of locality** and how it constrains the form of physical interactions and Lagrangians.
- Provide at least one example (e.g., Maxwells equations or Einsteins gravity) that illustrates how the principle of locality appears in physical laws.
- Describe briefly the general form of the action and Lagrangian density in classical field theory, and explain why only first-order time derivatives are usually included.

Problem 2. Free-Fall Motion in a Gravitational Field (20 points)

Consider a particle of mass m moving vertically under a uniform gravitational field g. The action is

$$S[z] = \int_{t_1}^{t_2} L\big(z, \dot{z}, t\big)\, dt, \qquad L(z, \dot{z}, t) = \tfrac{1}{2} m \dot{z}^{\,2} - mg\, z.$$

1. Find the conjugate momentum p and write down the Hamiltonian $H(z, p)$ for this system.

   Solution: The conjugate momentum is

   $$p \equiv \frac{\partial L}{\partial \dot{z}} = m\dot{z}.$$

   The Hamiltonian is

   $$H(z,p) \equiv p\dot{z} - L = p\left(\frac{p}{m}\right) - \left[\frac{1}{2} m \left(\frac{p}{m}\right)^2 - mg\, z\right] = \frac{p^2}{2m} + mg\, z.$$

2. Using the Euler-Lagrange equation, derive the equation of motion for $z(t)$.

   Solution: Euler-Lagrange equation:

   $$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{z}}\right) - \frac{\partial L}{\partial z} = 0 \quad \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}(m\dot{z}) - (-mg) = 0 \quad \Rightarrow \quad m\ddot{z} + mg = 0.$$

   Hence the equation of motion is

   $$\ddot{z} = -\,g.$$

3. Solve the equation of motion and express $z(t)$ in terms of the initial position $z(0)$ and initial velocity $\dot{z}(0)$.

   Solution: Integrate twice and impose the initial data $z(0)$ and $\dot{z}(0)$:

   $$\dot{z}(t) = \dot{z}(0) - gt, \qquad z(t) = z(0) + \dot{z}(0)\, t - \tfrac{1}{2} g t^2,$$

   where $z(t)$ is the solution of the equation of motion, which also describes the trajectory of the particle.

Problem 3. Deriving Maxwells Equations from the Electromagnetic Action (20 points)

Consider the electromagnetic field in vacuum, described by the four-potential $A_\mu(x)$ and the field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The action of the electromagnetic field (in natural units $c = \hbar = 1$) is

$$S[A] = \int d^4x\, \mathcal{L}, \qquad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

1. By varying the action with respect to A_ν, derive the Euler-Lagrange equation

   $$\partial_\mu F^{\mu\nu} = 0,$$

   which corresponds to the **inhomogeneous Maxwell equations** in vacuum.

   Solution: We vary the action with respect to A_ν and assume δA_ν vanishes on the boundary:

   $$S[A] = \int d^4x\, \mathcal{L}, \qquad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

   Compute δS:

   $$\delta S = \int d^4x \left(-\tfrac{1}{4}\, \delta(F_{\mu\nu} F^{\mu\nu}) \right) = \int d^4x \left(-\tfrac{1}{2}\, F^{\mu\nu}\, \delta F_{\mu\nu} \right),$$

   using

   $$\begin{aligned} \delta(F_{\mu\nu} F^{\mu\nu}) &= \delta F^{\mu\nu} F_{\mu\nu} + F^{\mu\nu} \delta F_{\mu\nu} = \delta F^{\mu\nu} F_{\mu\nu} + g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} \delta F_{\rho\sigma} \\ &= \delta F^{\mu\nu} F_{\mu\nu} + F^{\rho\sigma} \delta F_{\rho\sigma} = 2F^{\mu\nu} \delta F_{\mu\nu}. \end{aligned}$$

   Now

   $$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu,$$

   so

   $$\delta S = -\frac{1}{2} \int d^4x\, F^{\mu\nu} \big(\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu \big) = - \int d^4x\, F^{\mu\nu}\, \partial_\mu \delta A_\nu,$$

   where antisymmetry $F^{\mu\nu} = -F^{\nu\mu}$ makes the two terms equal and gives the factor of 2. Integrate by parts and drop the boundary term:

   $$\delta S = \int d^4x\, (\partial_\mu F^{\mu\nu})\, \delta A_\nu.$$

Since δA_ν is arbitrary in the bulk, stationarity $\delta S = 0$ implies

$$\boxed{\partial_\mu F^{\mu\nu} = 0}.$$

2. The tensor $F_{\mu\nu}$ is antisymmetric by definition:

$$F_{\mu\nu} = -F_{\nu\mu}.$$

Using this property, and assuming that the partial derivatives commute ($\partial_\alpha \partial_\beta = \partial_\beta \partial_\alpha$), show that $F_{\mu\nu}$ satisfies

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

This identity represents the **homogeneous Maxwell equations**.

Solution: By definition,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \Rightarrow \quad F_{\mu\nu} = -F_{\nu\mu}.$$

Compute the cyclic sum:

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = \partial_\alpha(\partial_\beta A_\gamma - \partial_\gamma A_\beta) + \partial_\beta(\partial_\gamma A_\alpha - \partial_\alpha A_\gamma) + \partial_\gamma(\partial_\alpha A_\beta - \partial_\beta A_\alpha).$$

Group like terms:

$$\big(\partial_\alpha \partial_\beta A_\gamma - \partial_\beta \partial_\alpha A_\gamma\big) + \big(\partial_\beta \partial_\gamma A_\alpha - \partial_\gamma \partial_\beta A_\alpha\big) + \big(\partial_\gamma \partial_\alpha A_\beta - \partial_\alpha \partial_\gamma A_\beta\big).$$

Assuming partial derivatives commute, $\partial_\mu \partial_\nu = \partial_\nu \partial_\mu$, each parenthesis vanishes. Hence

$$\boxed{\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0}$$

which is the homogeneous Maxwell identity (Bianchi identity).

3. Using the identifications

$$E_i = F_{i0}, \qquad B^i = \tfrac{1}{2}\varepsilon^{ijk} F_{jk},$$

express the four-dimensional equations from parts (1) and (2) in 3+1 form, and verify that they reproduce the familiar vacuum Maxwell equations:

$$\nabla\cdot\mathbf{E}=0,\qquad \nabla\cdot\mathbf{B}=0,\qquad \nabla\times\mathbf{E}+\partial_t\mathbf{B}=0,\qquad \nabla\times\mathbf{B}-\partial_t\mathbf{E}=0.$$

Solution: From part (1) we have the inhomogeneous Maxwell equations in vacuum:

$$\partial_\mu F^{\mu\nu}=0.$$

Split by components. the covariant components of $F_{\mu\nu}$ is

$$F_{\mu\nu}=\begin{pmatrix}0 & E_x & E_y & E_z\\ E_x & 0 & B_z & -B_y\\ E_y & -B_z & 0 & B_x\\ E_z & B_y & -B_x & 0\end{pmatrix}.$$

Gauss for $\mathbf{E}$: set $\nu=0$:

$$0=\partial_\mu F^{\mu 0}=\partial_i F^{i0}=\partial_i(-E_i)\ \Rightarrow\ \nabla\cdot\mathbf{E}=0.$$

Ampère - Maxwell (no sources): set $\nu=j$:

$$0=\partial_\mu F^{\mu j}=\partial_0 F^{0j}+\partial_i F^{ij}=\partial_t E_j+\partial_i\left(-\varepsilon^{ijk}B_k\right)=\partial_t E_j-(\nabla\times\mathbf{B})_j,$$

so

$$\nabla\times\mathbf{B}-\partial_t\mathbf{E}=0.$$

From part (2) we have the homogeneous identity (Bianchi identity)

$$\partial_\alpha F_{\beta\gamma}+\partial_\beta F_{\gamma\alpha}+\partial_\gamma F_{\alpha\beta}=0.$$

Again split by components.
Gauss for $\mathbf{B}$: take purely spatial indices $(\alpha,\beta,\gamma)=(i,j,k)$:

$$0=\partial_i F_{jk}+\partial_j F_{ki}+\partial_k F_{ij}=\partial_i(\varepsilon_{jk\ell}B^\ell)+\text{cyc}=\varepsilon_{jk\ell}\,\partial_i B^\ell+\text{cyc}\ \Rightarrow\ \nabla\cdot\mathbf{B}=0.$$

(Contract with ε^{ink} and use $\varepsilon^{ijk}\varepsilon_{jk\ell}=2\delta^i{}_\ell$.)
Faradays law: take one time and two space indices, e.g. $(\alpha,\beta,\gamma)=(0,i,j)$:

$$0=\partial_0 F_{ij}+\partial_i F_{j0}+\partial_j F_{0i}=\partial_t(\varepsilon_{ijk}B^k)-\partial_i E_j+\partial_j E_i.$$

Contract with $\frac{1}{2}\varepsilon^{\ell ij}$ to get

$$(\partial_t \mathbf{B})^\ell + (\nabla \times \mathbf{E})^\ell = 0 \quad \Rightarrow \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0.$$

Collecting results:

$$\boxed{\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad \nabla \times \mathbf{B} - \partial_t \mathbf{E} = 0}$$

which are the familiar vacuum Maxwell equations in $3+1$ form.

4. Briefly explain why the action $S[A]$ represents a **local and Lorentz-invariant** field theory.

Solution: The action

$$S[A] = \int d^4x \, \mathcal{L}, \qquad \mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

is both **local** and **Lorentz invariant** because:

- *Locality:* The Lagrangian density $\mathcal{L}$ depends only on the field $A_\mu(x)$ and its derivatives $\partial_\mu A_\nu(x)$ evaluated at the same spacetime point x. There are no nonlocal terms such as integrals or functions of fields at different points $x' \neq x$. Hence, interactions are local.
- *Lorentz invariance:* The field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ transforms covariantly under Lorentz transformations:

$$F'_{\mu\nu} = \Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma F_{\rho\sigma}.$$

Therefore, the scalar contraction $F_{\mu\nu} F^{\mu\nu}$ is invariant. Since the measure d^4x is also Lorentz invariant, the whole action $S[A]$ remains unchanged under Lorentz transformations.

Hence, $S[A]$ describes a **local, Lorentz-invariant field theory** for the electromagnetic field.